

Digital Material Passport

ID DMP-2024-ARR-001 - Version 0.1.0
Issue Date 2024-12-15 - Certificate Type EN 10204 3.1

Manufacturer	Customer	Business Transaction	
Advanced Steel Technologies GmbH Industriestraße 15 52070 Aachen DE quality@advanced-steel.de	Precision Engineering Ltd. Manufacturing Park 42 Birmingham B1 1AA GB procurement@precision-eng.co.uk	Order	PO-2024-7890
		Delivery	DN-2024-3456 - 500 kg

Product

Product Name	Hardenability Test Steel	Shape	RoundBar - D 100 mm - Length 1000 mm
Batch ID	HTB-2024-045		

Chemical Analysis

Heat Number H-2024-078 - Sample Location Ladle

Symbol	C	Mn	Si	P	S	Cr	Ni
Unit	%	%	%	%	%	%	%
Min	0.42	0.5					
Max	0.5	0.8	0.4	0.03	0.025	0.25	0.25
Actual	0.45	0.65	0.25	0.015	0.008	0.12	0.08

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Jominy Hardenability Test					ISO 377	✓
Quenched from 850°C						

Distance from - quenched end (mm)	1.5	3	5	7	9	11	15	20	25	30
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Min	42	39	35	32	29	26	22	20		
Max	47	46	44	41	39	37	33	31	<= 30	<= 29
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile	ISO 2178	✓
Magnetic induction method		

Position along length (mm)	0	50	100
Value [µm]	125	115	108
Min	>= 120	>= 110	>= 100
Max	<= 130	<= 125	<= 120
Status	Pass	Pass	Pass

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer,
Quality Control Laboratory - 2024-12-15